I. Optimal Scheduling Problem formulation

This section introduces the mixed integer optimization program we solve, with the goal of determining the optimal locations, timing, and charging/discharging rates for MCSs; along with optimizing the working schedule of the CEVs. The addition of prediction of working schedule of CEVs is motivated by the results in Table **??** elucidating that the subactivity powers of the CEV can vary; and thus their optimal scheduling is crucial to determine when they will charge from which MCS, which in turn determines the impact MCS charging has on the grid.

To meet the CEV (denoted by set $\mathcal{E}$) charging demands in a specific area, a network consisting of MCSs (denoted by set $\mathcal{M}$), depots for MCS charging called grid connection nodes (denoted by set $\mathcal{N}^g$), and construction nodes (denoted by set $\mathcal{N}^c$) is proposed. The MCSs are towed by electric pickups from one node to another, and the CEVs, which are assigned to fixed construction nodes (to account for their general immobility), are charged through the MCSs. For simplicity, we assume that the electric pickups are themselves powered by the MCS.

Note that in the following sections, $\mathcal{T}$ is the set of operating time intervals which is used to time-index the control variables which are applied for an interval (like charging/discharging powers). For example, for a day-ahead (24 hour) scheduling horizon with a 15-min granularity starting at 00:00 h, $\mathcal{T} = \{1, 2, 3, \dots, t_{\text{end}}\}$, with $t = 1 \in \mathcal{T}$ representing the interval between 00:00-00:15 h; $t = 0$ denotes the initial time instant 00:00 h.

A. Objective Function

The objective is to minimize the total operating cost over the scheduling horizon, and is formulated as,

$$
\begin{aligned}
\underset{\mathcal{D}}{\text{minimize}}\ J = & \sum_{m\in\mathcal{M}}\sum_{t\in\mathcal{T}} \lambda_t^{\text{elec}} P_{m,t}^{\text{ch,tot}} \Delta T \\
& + \sum_{m\in\mathcal{M}}\sum_{t\in\mathcal{T}} \lambda_t^{\text{CO}_2} \lambda^{\text{em}} P_{m,t}^{\text{ch,tot}} \Delta T \\
& + \lambda^{\text{NC}} P^{\text{NC}} + \lambda^{\text{OP}} P^{\text{OP}} \\
& + \rho^{\text{miss}} \sum_{i\in\mathcal{N}^c}\sum_{a\in\mathcal{A}} s_{i,a}^{\text{miss}} \\
& + \rho^{\text{travel}} \Delta T \sum_{m\in\mathcal{M}}\sum_{i,j\in\mathcal{N}}\sum_{t\in\mathcal{T}} y_{m,i,j,t}. \qquad (1)
\end{aligned}
$$

The first term in (1) represents the energy cost of charging the MCSs from the grid; with λ_t^{elec} and $P_{m,t}^{\text{ch,tot}}$ denoting the electricity charge rate (\$/kWh) and total charging power (kW) of MCS m from the grid over time interval t; respectively. The second term in (1) represents the monetized carbon-emissions cost associated with grid energy consumption; with $\lambda_t^{\text{CO}_2}$ and λ^{em} denoting the carbon emissions intensity (kgCO_2/kWh) and conversion factor of carbon emissions to dollar terms (\$/kgCO_2); respectively. Δt represents the granularity of the scheduling horizon in hours. The third and fourth term in (1) captures the non-coincident (NC) and on-peak (OP) demand charges. NC demand charges (NCDC) are the cost levied for the maximum power withdrawn from the grid in a month, while OP demand charges (OPDC) are the cost levied for the maximum power withdrawn from the grid in all the on-peak hours (4-9 pm) of the month. λ^{NC} and λ^{OP} denote the NCDC rate and OPDC rate (\$/kW); respectively, while $P^{\text{NC}} = \max\{\sum_{m\in\mathcal{M}} P_{m,t}^{\text{ch,tot}}\}_{t\in\mathcal{T}}$ and $P^{\text{OP}} = \max\{\sum_{m\in\mathcal{M}} P_{m,t}^{\text{ch,tot}}\}_{t\in\mathcal{T}^{\text{op}}}$ denotes the NC demand peak and OP demand peak; respectively.[1] The fifth term in (1) penalizes unmet construction work requirements, where $s_{i,a}^{\text{miss}}$ is the missed work duration (hours) for subactivity a at construction node i over the entire scheduling horizon. ρ^{miss} is a conversion factor of missed work to dollar terms (\$/hour), based on construction crew labor, delay, and operational costs. The last term in (1) penalizes the labor cost for towing the MCSs' from one place to another over the scheduling horizon; where $y_{m,i,j,t}$ is a binary variable which takes a value of 1 if the MCS m is moving along path (i, j) over time interval t, and is 0 otherwise, and ρ^{travel} is the labor cost (\$/hour) associated with towing the MCS from one place to another. This discourages excessive MCS relocation and promotes spatially efficient charging schedules.

$\mathcal{D} = \Big\{P_{m,i,t}^{\text{ch,MCS}}, P_{m,i,t}^{\text{dch,MCS}}, P_{m,i,e,t}^{\text{MCS}\to\text{CEV}}, P_{i,e,t}^{\text{work}}, s_{i,a}^{\text{miss}}, u_{i,e,t,a}, \rho_{m,i,e,t}, \beta_{m,i,t}^{\text{arr}}, \beta_{m,i,t}^{\text{dep}}, x_{m,i,j,t}, , y_{m,i,j,t}, \mu_{i,e,t}, z_{m,i,t}, g_{m,i,t}\Big\}$ represents the set of all the decision variables of the optimization, and for space limitations not explicitly written in (1). Note that not all terms in $\mathcal{D}$ find representation in the objective function, and the reader is requested to refer to the nomenclature or the constraints below for a clearer exposition.

B. Constraints

In this section, we present the constraints of the optimization algorithm. Because of the extensiveness of the constraints, we divide it into four subsections as presented below.

1) Charging and discharging power constraints of the MCSs and CEVs:

$$
P_{m,t}^{\text{ch,tot}} = \sum_{i\in\mathcal{N}^g} P_{m,i,t}^{\text{ch,MCS}}, \quad \forall m\in\mathcal{M},\ t\in\mathcal{T}, \qquad (2a)
$$

$$
P_{m,t}^{\text{dch,tot}} = \sum_{i\in\mathcal{N}^c} P_{m,i,t}^{\text{dch,MCS}}, \quad \forall m\in\mathcal{M},\ t\in\mathcal{T}, \qquad (2b)
$$

$$
P_{m,i,t}^{\text{dch,MCS}} = 0, \quad \forall m\in\mathcal{M},\ i\in\mathcal{N}^g,\ t\in\mathcal{T}, \qquad (2c)
$$

$$
P_{m,i,t}^{\text{ch,MCS}} = 0, \quad \forall m\in\mathcal{M},\ i\in\mathcal{N}^c,\ t\in\mathcal{T}, \qquad (2d)
$$

$$
P_{m,i,t}^{\text{dch,MCS}} = \sum_{e\in\mathcal{E}} P_{m,i,e,t}^{\text{MCS}\to\text{CEV}}, \quad \forall m\in\mathcal{M},\ i\in\mathcal{N}^c,\ t\in\mathcal{T}. \qquad (2e)
$$

$$
P_{m,i,t}^{\text{ch,MCS}}, P_{m,i,t}^{\text{dch,MCS}}, P_{m,i,e,t}^{\text{MCS}\to\text{CEV}} \geq 0, \quad \forall m\in\mathcal{M},\ i\in\mathcal{N}^c,\ t\in\mathcal{T}. \qquad (2f)
$$

Constraints (2a) and (2b) define the total charging and discharging powers of each MCS m over time interval t as a sum of node specific charging and discharging powers over all grid and construction nodes; respectively. Constraints (2c)

[1] Although for electricity bills, the NC demand peak and OP demand peak are computed over a month, in operation it is rare to schedule CEV/MCS over a month. Thus, in practical scenarios we penalize the peaks over the scheduling horizon with the intuition that repeatedly minimizing the peaks over rolling scheduling horizons acts as a surrogate to minimize the monthly peaks [**?**].

and (2d) enforce that MCSs charge only at grid nodes and discharge only at construction nodes; respectively. Constraint (2e) ensures that the discharging power of an MCS m at construction node i over time interval t equals the total power transferred from that MCS to all the connected CEVs, where $P^{\mathrm{MCS}\to\mathrm{CEV}}_{m,i,e,t}$ denotes the power transferred by MCS m, at node i, to CEV e over time interval t. Constraint (2f) ensures that all the charging and discharging powers of the MCS, and the power transfer from the MCS to CEVs are non-negative.

$$P^{\mathrm{ch,MCS}}_{m,i,t} \leq CH^{\mathrm{MCS}}_{m} g_{m,i,t},\ \forall m \in \mathcal{M},\ i \in \mathcal{N}^{g},\ t \in \mathcal{T}, \tag{3a}$$

$$g_{m,i,t} \leq z_{m,i,t},\ \ \forall m \in \mathcal{M},\ i \in \mathcal{N}^{g},\ t \in \mathcal{T}, \tag{3b}$$

$$\sum_{m \in \mathcal{M}} g_{m,i,t} \leq 1,\ \ \forall i \in \mathcal{N}^{g},\ t \in \mathcal{T}, \tag{3c}$$

$$P^{\mathrm{dch,MCS}}_{m,i,t} \leq DCH^{\mathrm{MCS}}_{m} z_{m,i,t},\ \forall m \in \mathcal{M},\ i \in \mathcal{N}^{c},\ t \in \mathcal{T}. \tag{3d}$$

Here $z_{m,i,t}$ and $g_{m,i,t}$ are binary variables: $z_{m,i,t} = 1$ if and only if (iff) MCS m is present at node i over interval t, and $g_{m,i,t} = 1$ iff MCS m is drawing charge from the grid at node i over interval t (both 0 otherwise). Constraint (3a) permits grid charging only when $g_{m,i,t} = 1$ and caps it at the MCS charging capacity CH^{MCS}_{m}; (3b) allows an MCS to charge only at a node where it is physically present ($g_{m,i,t} \leq z_{m,i,t}$); and (3c) restricts each grid-connection node to charging at most one MCS per interval, which might give rise to the charging congestion. Constraint (3d) analogously limits the discharging power at each construction node to the discharging capacity DCH^{MCS}_{m}, permitted only when the MCS is present there ($z_{m,i,t} = 1$).

$$P^{\mathrm{MCS}\to\mathrm{CEV}}_{m,i,e,t} \leq DCH^{\mathrm{plug}}_{m} \rho_{m,i,e,t},\qquad \forall m \in \mathcal{M},\ i \in \mathcal{N}^{c},\ e \in \mathcal{E},\ t \in \mathcal{T}, \tag{4a}$$

$$\sum_{m \in \mathcal{M}} P^{\mathrm{MCS}\to\mathrm{CEV}}_{m,i,e,t} \leq CH^{\mathrm{CEV}}_{e} \mu_{i,e,t}, \forall i \in \mathcal{N}^{c},\ e \in \mathcal{E},\ t \in \mathcal{T}, \tag{4b}$$

$$\sum_{e \in \mathcal{E}} \rho_{m,i,e,t} \leq C^{\mathrm{plug}}_{m},\quad \forall m \in \mathcal{M},\ i \in \mathcal{N}^{c},\ t \in \mathcal{T}. \tag{4c}$$

Constraint (4a) limits the power delivered through each MCS plug and forces the MCS-to-CEV power to zero unless the binary connection variable $\rho_{m,i,e,t}$ denoting the connection status of MCS m at node i to CEV e over time interval t is 1. Constraint (4b) enforces the charging acceptance rate of each CEV by $CH^{\mathrm{CEV}}_{e}$and permits charging iff the binary variable $\mu_{i,e,t} = 1$, indicating the CEV e is ready to accept charge at node i over time interval t. Note that the binary variable $\mu_{i,e,t}$ is independent of which MCS charges the CEV e; however the binary variable $\rho_{m,i,e,t}$ is linked to which MCS m charges CEV e. Constraint (4c) limits the number of CEVs simultaneously connected to a given MCS by the number of outlet plugs from the MCS, given by C^{plug}_{m}.

2) Energy consumption and state-of-energy (SOE) constraints of the MCSs and CEVs:

$$0 \leq P^{\mathrm{work}}_{i,e,t} \leq R_{e,t} A_{i,e} \left(1 - \mu_{i,e,t}\right), \forall i \in \mathcal{N}^{c},\ e \in \mathcal{E},\ t \in \mathcal{T}, \tag{5a}$$

$$R_{e,t} = R^{\mathrm{work}}_{e,t},\qquad \forall e \in \mathcal{E},\ t \in \mathcal{T}^{\mathrm{work}} \tag{5b}$$

$$R_{e,t} = 0,\qquad \forall e \in \mathcal{E},\ t \in \mathcal{T} \setminus \{\mathcal{T}^{\mathrm{work}}\} \tag{5c}$$

$$\sum_{a \in \mathcal{A}} u_{i,e,t,a} + \mu_{i,e,t} \leq 1, \forall i \in \mathcal{N}^{c},\ e \in \mathcal{E},\ t \in \mathcal{T}, \tag{5d}$$

$$P^{\mathrm{work}}_{i,e,t} = \sum_{a \in \mathcal{A}} p_a u_{i,e,t,a}, \forall i \in \mathcal{N}^{c},\ e \in \mathcal{E},\ t \in \mathcal{T}. \tag{5e}$$

The above constraints couple the charging decisions of the CEVs with their work schedule and construction node assignment. Constraint (5a) limit the work power $P^{\mathrm{work}}_{i,e,t}$, of CEV e at node i over time interval t by the CEV work capacity $R_{e,t}$, the assignment matrix $A_{i,e}$, and the charging connection status $\mu_{i,e,t}$. $R_{e,t}$ takes a value of working capacity (which is a manufacturer specification) of the CEV e during work hours and is 0 for other hours as specified in (5c), while $A_{i,e}$ takes a value of 1 iff CEV e is assigned to node i, and is 0 otherwise. Note that the power consumption of CEVs is constrained to be non-negative in (5a). Constraint (5d) enforces that a CEV cannot charge and work over the same time interval t, where $u_{i,e,t,a}$ is a binary variable denoting 1 iff CEV e is performing subactivity a at node i over time interval t. Constraint (5e) defines the work power from the selected construction subactivity, where p_a is the power consumption of the CEV while performing subactivity a.

$$L^{\mathrm{trv}}_{m,i,j,t} = k^{\mathrm{trv}}_{m} \Delta T y_{m,i,j,t},\qquad \forall m \in \mathcal{M},\ i,j \in \mathcal{N},\ t \in \mathcal{T}, \tag{6a}$$

$$L^{\mathrm{trv,tot}}_{m,t} = \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} L^{\mathrm{trv}}_{m,i,j,t},\quad \forall m \in \mathcal{M},\ t \in \mathcal{T}. \tag{6b}$$

Constraint (6a) represents the travel energy consumption $L^{\mathrm{trv}}_{m,i,j,t}$ (kWh) when the MCS m is moving along the path (i,j) from node i to node j over time interval t. Here k^{trv}_{m} denotes the energy consumed per hour by the MCS m.[2] The binary variable $y_{m,i,j,t}$ indicates the travel status of MCS m from along path (i,j) over time interval t. Constraint (6b) is the total energy consumption of the MCS m over time interval t, and is expressed by summing (6a) over all paths.

$$\mathrm{SOE}^{\mathrm{MCS}}_{m,t+1} = \mathrm{SOE}^{\mathrm{MCS}}_{m,t} + \eta_m P^{\mathrm{ch,tot}}_{m,t} \Delta T - \frac{P^{\mathrm{dch,tot}}_{m,t} \Delta T}{\eta_m} - L^{\mathrm{trv,tot}}_{m,t},\qquad \forall m \in \mathcal{M},\ t \in \mathcal{T}, \tag{7a}$$

$$\mathrm{SOE}^{\mathrm{CEV}}_{e,t+1} = \mathrm{SOE}^{\mathrm{CEV}}_{e,t} + \sum_{m \in \mathcal{M}} \sum_{i \in \mathcal{N}^{c}} P^{\mathrm{MCS}\to\mathrm{CEV}}_{m,i,e,t} \Delta T - \sum_{i \in \mathcal{N}^{c}} P^{\mathrm{work}}_{i,e,t} \Delta T,\quad \forall e \in \mathcal{E},\ t \in \mathcal{T}. \tag{7b}$$

[2]Here, we assume that the electric pickup has a manufacturer specified energy consumption while traveling at different speeds. Thus, for all the paths (i,j), the user can pre-decide the speed (assumed constant), and thus the energy consumption, k^{trv}_{m}, of the electric pickup.

Equation (7a) updates the MCS SOE by adding grid-charged energy, subtracting MCS-to-CEV discharged energy after efficiency losses, and subtracting travel energy. Equation (7b) updates the CEV SOE by adding energy received from MCSs and subtracting energy consumed for work.

$$\mathrm{SOE}^{\mathrm{MCS}}_{m,0} = \mathrm{SOE}^{\mathrm{MCS,ini}}_{m} = \mathrm{SOE}^{\mathrm{MCS}}_{m,t_{\mathrm{end}}}, \quad \forall m \in \mathcal{M}, \tag{8a}$$

$$\mathrm{SOE}^{\mathrm{CEV}}_{e,0} = \mathrm{SOE}^{\mathrm{CEV,ini}}_{e} = \mathrm{SOE}^{\mathrm{CEV}}_{e.t_{\mathrm{end}}}, \quad \forall e \in \mathcal{E}, \tag{8b}$$

$$\mathrm{SOE}^{\mathrm{MCS,min}}_{m} \le \mathrm{SOE}^{\mathrm{MCS}}_{m,t} \le \mathrm{SOE}^{\mathrm{MCS,max}}_{m}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}, \tag{8c}$$

$$\mathrm{SOE}^{\mathrm{CEV,min}}_{e} \le \mathrm{SOE}^{\mathrm{CEV}}_{e,t} \le \mathrm{SOE}^{\mathrm{CEV,max}}_{e}, \quad \forall e \in \mathcal{E},\ t \in \mathcal{T}. \tag{8d}$$

Equations (8a)–(8b) set the initial and final SOE of all MCSs and CEVs to be equal to impose an energy-neutral terminal condition, preventing the optimizer from artificially depleting batteries at the end of the horizon, and be ready for next day's work. Constraints (8c)–(8d) enforce the allowable SOE operating ranges for the MCS and CEV (see Table **??**); respectively.

3) Spatial routing and connection status of the MCSs and CEVs:

$$\rho_{m,i,e,t} \le A_{i,e}, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ e \in \mathcal{E},\ t \in \mathcal{T}, \tag{9a}$$

$$\rho_{m,i,e,t} \le z_{m,i,t}, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ e \in \mathcal{E},\ t \in \mathcal{T}. \tag{9b}$$

Constraint (9a) allows CEV e to connect at node i iff the CEV is assigned to that node, through the assignment matrix. Similarly, constraint (9b) allows a CEV to connect to MCS m at node i iff the MCS is physically present at that node.

$$x_{m,i,i,t} = 0, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ t \in \mathcal{T}, \tag{10a}$$

$$y_{m,i,j,t} = \sum_{\tau=\max\{1,\ t-\tau^{\mathrm{trv}}_{i,j}+1\}}^{t} x_{m,i,j,\tau}, \quad \forall m \in \mathcal{M},\ i,j \in \mathcal{N},\ i \neq j,\ t \in \mathcal{T} \tag{10b}$$

$$\sum_{i\in\mathcal{N}} z_{m,i,t} + \sum_{\substack{i,j\in\mathcal{N}\\ i\neq j}} y_{m,i,j,t} = 1, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}, \tag{10c}$$

$$\sum_{i\in\mathcal{N}^g} z_{m,i,1} + \sum_{i\in\mathcal{N}^g}\sum_{\substack{j\in\mathcal{N}\\ j\neq i}} x_{m,i,j,1} = 1, \quad \forall m \in \mathcal{M}, \tag{10d}$$

$$\sum_{i\in\mathcal{N}^g} z_{m,i,t_{\mathrm{end}}} = 1, \quad \forall m \in \mathcal{M}. \tag{10e}$$

$x_{m,i,j,t}$ is a binary variable which takes a value of 1 iff MCS m departs node i towards node j (along path (i,j)) over time interval t. Constraint (10a) eliminates travel from a node to itself. Constraint (10b) defines the binary travel status indicator of MCS m over time interval t along path (i,j) using the departure decision. Specifically, an MCS is in transit on path (i,j) over time interval t if it departed from node i toward node j within the previous $\tau^{\mathrm{trv}}_{i,j}$ intervals. $\tau^{\mathrm{trv}}_{i,j}$ is an user input based on the network topology which denotes the time-steps (under the discretization ΔT) it takes for an MCS to travel from node i to node j along path (i,j).[3] Constraint (10c) ensures that an MCS is either parked at one node or traveling on one route during a time interval. Constraints (10d)–(10e) require each MCS to start from a grid node and end the horizon parked at a grid node.

$$\beta^{\mathrm{dep}}_{m,i,t} = \sum_{\substack{j\in\mathcal{N}\\ j\neq i}} x_{m,i,j,t}, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ t \in \mathcal{T}, \tag{11a}$$

$$\beta^{\mathrm{arr}}_{m,j,t} = \sum_{\substack{i\in\mathcal{N}\\ i\neq j\\ t-\tau^{\mathrm{trv}}_{i,j}\in\mathcal{T}}} x_{m,i,j,t-\tau^{\mathrm{trv}}_{i,j}}, \quad \forall m \in \mathcal{M},\ j \in \mathcal{N},\ t \in \mathcal{T}, \tag{11b}$$

$$\beta^{\mathrm{arr}}_{m,i,t} - \beta^{\mathrm{dep}}_{m,i,t} = z_{m,i,t} - z_{m,i,t-1}, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ t \in \mathcal{T}\setminus\{1\}, \tag{11c}$$

$$\sum_{t\in\mathcal{T}} \beta^{\mathrm{arr}}_{m,i,t} = \sum_{t\in\mathcal{T}} \beta^{\mathrm{dep}}_{m,i,t}, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N}, \tag{11d}$$

$$\beta^{\mathrm{arr}}_{m,i,t} + \beta^{\mathrm{dep}}_{m,i,t} \le 1, \quad \forall m \in \mathcal{M},\ i \in \mathcal{N},\ t \in \mathcal{T}. \tag{11e}$$

Constraint (11a) defines the binary departure indicator of an MCS, $\beta^{\mathrm{dep}}_{m,i,t}$, which takes a value of 1 iff MCS m departs node i along some outgoing node $j \neq i$ over time interval t. Constraint (11b) accounts for travel time by linking the binary arrival indicator of an MCS, $\beta^{\mathrm{arr}}_{m,j,t}$, which takes a value of 1 iff MCS m arrives at node j over time interval t from some incoming node i through a path that started $\tau^{\mathrm{trv}}_{i,j}$ intervals earlier. Constraint (11c) updates the MCS node-presence variable $z_{m,i,t}$ using arrivals and departures. Constraint (11d) balances total arrivals and departures over the horizon, while (11e) prevents an MCS from arriving at and departing from the same node in the same interval.

4) Operational adjustment for CEV work scheduling: Here, we present a distinction between all the CEV subactivities in $\mathcal{A}$, and productive construction related subactivities (only digging, and loading+swinging), in $\mathcal{A}^{\mathrm{con}} \subseteq \mathcal{A}$, to model the CEV traveling. That is because, CEV travel is not a productive subactivity in itself; but even within a single construction node, the CEV cannot work a single spot indefinitely, and must periodically travel (reposition its undercarriage) to reach fresh material and dumping locations.

[3] As it may be challenging to understand the interaction between the binary variables $z_{m,i,t}$, $x_{m,i,j,t}$, $y_{m,i,j,t}$, $\beta^{\mathrm{dep}}_{m,i,t}$ and $\beta^{\mathrm{arr}}_{m,j,t}$, and the integer user-input $\tau^{\mathrm{trv}}_{i,j}$ to model the travel time delay, we present a representative example in Appendix B in an extended version of the paper [].

$$\sum_{a\in\mathcal{A}} u_{i,e,t,a} \le 1, \qquad \forall i \in \mathcal{N}^c, e \in \mathcal{E},\ t \in \mathcal{T}, \tag{12a}$$

$$u_{i,e,t,a} \le A_{i,e}, \qquad \forall e \in \mathcal{E},\ i \in \mathcal{N}^c,\ a \in \mathcal{A},\ t \in \mathcal{T}, \tag{12b}$$

$$\Delta T \sum_{e\in\mathcal{E}} \sum_{t\in\mathcal{T}} u_{i,e,t,a} + s^{\text{miss}}_{i,a} = H_{i,a},\ \forall i \in \mathcal{N}^c,\ a \in \mathcal{A}^{\text{con}}, \tag{12c}$$

$$\gamma_{a\to a'} \sum_{e\in\mathcal{E}} \sum_{\tau=1}^{t} u_{i,e,\tau,a'} \le \kappa^{\text{seq}}_{a\to a'} \sum_{e\in\mathcal{E}} \sum_{\tau=1}^{t} u_{i,e,\tau,a} \gamma_{a\to a'},$$
$$\forall i \in \mathcal{N}^c,\ a, a' \in \mathcal{A},\ \ t \in \mathcal{T}, \tag{12d}$$

$$\sum_{\tau=t}^{t+t_{\text{limit}}/\Delta T} \sum_{a\in\mathcal{A}} u_{i,e,\tau,a} \le \frac{t_{\text{limit}}}{\Delta T},$$
$$\forall i \in \mathcal{N}^c, e \in \mathcal{E},\ t \in \{1, 2, 3, \dots, t_{\text{end}} - \frac{t_{\text{limit}}}{\Delta T}\}. \tag{12e}$$

Constraint (12a) allows each CEV to perform at most one subactivity over each time interval at each site. Constraint (12b) restricts activity assignment to the CEV's construction node assignment. Equation (12c) enforces the required productive work duration $H_{i,a}$ at each site which is an user input, with slack $s^{\text{miss}}_{i,a}$ capturing unmet work in hours. Constraint (12d) imposes the cumulative precedence relation between subactivities at a site, e.g., by time t, cumulative duration of loading+swinging cannot exceed the allowed multiple of cumulative duration of digging, dented by $\kappa^{\text{seq}} \in \mathbb{N}$. This prevents loading+swinging activities from being scheduled before sufficient digging has been completed, thereby ensuring the physical consistency of the construction workflow. $\gamma_{a\to a'}$ is a user input which is 1 if there is a precedence relationship between activities a to a'; otherwise it is 0. Constraint (12e) imposes an operational rest requirement on each CEV. Specifically, over any $t_{\text{limit}} + \Delta T$ consecutive times (in hours), a CEV is allowed to perform construction related work (digging, loading+swinging, or traveling) in at most t_{limit} hours. Therefore, each CEV must remain idle or be unavailable for work in at least one interval within every rolling $t_{\text{limit}} + 1$ interval window. This constraint prevents uninterrupted operation over long periods and captures practical limitations associated with operator breaks, equipment repositioning and preventing overheating, or other operational interruptions.

$$\kappa^{\text{wt}} \sum_{\tau=1}^{t} u_{i,e,\tau,a^{\text{trv}}} \le \sum_{a\in\mathcal{A}^{\text{con}}} \sum_{\tau=1}^{t} u_{i,e,\tau,a},$$
$$i \in \mathcal{N}^c, \forall e \in \mathcal{E},\ t \in \mathcal{T}, \tag{13a}$$

$$\kappa^{\text{wt}} \sum_{\tau=1}^{t} u_{i,e,\tau,a^{\text{trv}}} \ge \sum_{a\in\mathcal{A}^{\text{con}}} \sum_{\tau=1}^{t} u_{i,e,\tau,a} - \kappa^{\text{wt}},$$
$$i \in \mathcal{N}^c, \forall e \in \mathcal{E},\ t \in \mathcal{T}. \tag{13b}$$

Constraints (13) couple the CEV's repositioning to its productive work, where a^{trv} denotes the traveling subactivity, and $\kappa^{\text{wt}} \in \mathbb{N}$ is the number of intervals of productive work between each travel interval. The upper bound (13a) prevents more than one travel for every κ^{wt} productive work intervals (avoiding spurious repositioning), while the lower bound (13b) forces at least one travel within every κ^{wt} productive work intervals (so work cannot proceed without repositioning). Together they yield an evenly interspersed productive work–travel pattern, consistent with the observed construction demonstration.